

Question 2: Parse Trees

$(a+(b*C)/2)$

Factor

└ (Expr)
└ Expr → Expr + Term
└ Expr → Term → Factor → identifier: a
└ Term → Term / Factor
└ Term → Factor
└ └ Factor → (Expr)
└ └ Expr → Term → Term * Factor
└ └ └ Term → Factor → identifier: b
└ └ └ Factor → identifier: C
└ Factor → number: 2

$a*(3+b)*4$

Expr → Term

└ Term → Term * Factor
└ Term → Term * Factor
└ └ Term → Factor → identifier: a
└ └ Factor → (Expr)
└ └ Expr → Expr + Term
└ └ Expr → Term → Factor → number: 3

| $\vdash$ Term $\rightarrow$ Factor $\rightarrow$ identifier: b

$\vdash$ Factor $\rightarrow$ number: 4

42c+3(a+b)

Expr $\rightarrow$ Expr + Term

| $\vdash$ Expr $\rightarrow$ Term $\rightarrow$ Term * Factor

| | $\vdash$ Term $\rightarrow$ Factor $\rightarrow$ number: 42

| $\vdash$ Factor $\rightarrow$ identifier: c

$\vdash$ Term $\rightarrow$ Term * Factor

| $\vdash$ Term $\rightarrow$ Factor $\rightarrow$ number: 3

$\vdash$ Factor $\rightarrow$ (Expr)

$\vdash$ Expr $\rightarrow$ Expr + Term

| $\vdash$ Expr $\rightarrow$ Term $\rightarrow$ Factor $\rightarrow$ identifier: a

$\vdash$ Term $\rightarrow$ Factor $\rightarrow$ identifier: b

Question 3: Extended Grammar (unary +/-)

Factor $\rightarrow$ Unary

Unary $\rightarrow$ "+" Unary

| "-" Unary

| Primary

Primary $\rightarrow$ "(" Expr ")"

| number

| identifier

Par tree for $(3+-3)*4$

Expr $\rightarrow$ Term

└─ Term $\rightarrow$ Term * Factor

└─┬─ Term $\rightarrow$ Factor $\rightarrow$ Unary

└─┬─┬─ Unary $\rightarrow$ Primary

└─┬─┬─┬─ Primary $\rightarrow$ (Expr)

└─┬─┬─┬─┬─ Expr $\rightarrow$ Expr + Term

└─┬─┬─┬─┬─┬─ Expr $\rightarrow$ Term $\rightarrow$ Factor $\rightarrow$ Unary $\rightarrow$ Primary $\rightarrow$ number: 3

└─┬─┬─┬─┬─┬─┬─ Term $\rightarrow$ Factor $\rightarrow$ Unary

└─┬─┬─┬─┬─┬─┬─┬─ Unary $\rightarrow$ "-" Unary

└─┬─┬─┬─┬─┬─┬─┬─┬─ Unary $\rightarrow$ Primary $\rightarrow$ number: 3

└─┬─┬─┬─┬─┬─┬─┬─┬─┬─ Factor $\rightarrow$ Unary $\rightarrow$ Primary $\rightarrow$ number: 4